Energy-dependent slow collective flavor conversion of supernova neutrinos

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We study collective slow flavor conversion (SFC) of supernova neutrinos with multi-energy, multi-angle simulations for three representative neutrino spectra in the early accretion, late accretion, and cooling phases, in which multiple crossings between the initial electron- and heavy-lepton-flavor spectra are present. By numerically solving the neutrino quantum kinetic equations in a local periodic box, we find that SFC triggered predominantly by the spatially inhomogeneous instabilities drives the system toward a spatially coarse-grained, quasi-stationary state, whose flavor conversion probability depends strongly on energy, angle, and the neutrino mass ordering. While we find that not all of the initial spectral crossings are completely erased in the final state, a simple, box-like analytical prescription inspired by studies of fast flavor conversions, which eliminates the spectral crossings, can reasonably approximate the post-SFC spectra. Using the initial and post-SFC spectra, we also evaluate the changes of the corresponding ν_e and $\bar{\nu}_e$ heating rates as well as the absorption equilibrium electron fraction (Y_e). Within the considered scenarios, we find that the heating rates are generally enhanced by up to $\sim 80\%$ due to the net conversion of ν_x to ν_e (and $\bar{\nu}_x$ to $\bar{\nu}_e$) above their crossing energy, provided that the energy spectra above the crossing energy differ substantially. For the absorption equilibrium Y_e, spectra changes due to SFC increase it by ~ 0.03 due to the relatively more enhanced ν_e absorption rate than $\bar{\nu}_e$, which potentially drives supernova materials to be more proton-rich. These results highlight the importance of energy-dependent treatments of SFC for supernova neutrinos.

I. INTRODUCTION

Dense neutrino gases in core-collapse supernovae (CCSNe) and binary neutron-star mergers can undergo collective flavor transformations driven by neutrino–neutrino coherent forward scattering [1–3]. Among the different manifestations of collective oscillations, slow flavor conversion (SFC) arises from the interplay between vacuum oscillations and neutrino self-interactions [4, 5]. By modifying the electron-flavor energy spectra, SFC can affect neutrino heating, the electron fraction Y_e of the ejecta, and the resulting heavy-element nucleosynthesis [6–8]. Due to the recently much-improved understanding of collective flavor oscillations, SFC has received renewed attention in studies of its instability properties [9–11], spatiotemporal evolution [12][1], and its relevance in realistic astrophysical settings [14–16].

In Ref. [12], we examined SFC in a single-energy, multi-angle system and demonstrated that its asymptotic state depends sensitively on the $\bar{\nu}_e/\nu_e$ number-density ratio and more modestly on the neutrino mass ordering. A single-energy treatment, however, cannot describe the different mean energies and spectral shapes of ν_e, $\bar{\nu}_e$, and ν_x, nor the resulting spectral crossings. These features allow different energy modes to experience different amounts of flavor conversion and are particularly important for charged-current rates of neutrinos with nucleons, whose energy weighting makes them sensitive to the high-energy tails of the post-SFC spectra.

In this work, we extend our previous study to a multi-energy, multi-angle system for three representative CCSN phases: early accretion, late accretion, and cooling. For both normal and inverted mass orderings, we perform the linear stability analysis and solve the nonlinear quantum kinetic equations in a one-dimensional periodic box until the system has reached a quasi-stationary state after being coarse-grained over the simulation volume. We then discuss the effect of SFC on neutrino energy spectra, quantify the changes in the ν_e and $\bar{\nu}_e$ absorption heating rates and in equilibrium Y_e, and test a phenomenological

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[1] We note that, during the completion of this manuscript, Ref. [13] which performed a similar study for SFC has appeared. Our work presented here has been done independently using a different simulation code. While the results of the two studies broadly agree, quantitative differences regarding the coarse-grained asymptotic state exist, which motivate different analytical approximations (see Secs. V and VII).

box prescription that estimates the asymptotic spectra directly from the initial lepton-number spectrum.

We find that SFC is predominantly triggered by spatially inhomogeneous instabilities. The coarse-grained, quasistationary flavor conversion probability in the final state depends on the energy, angle, and neutrino mass ordering. Interestingly, unlike in the single-energy system studied in Ref. [12], the initial spectral crossings are not completely erased in the coarse-grained, angle-averaged final state. Nevertheless, it can still be reasonably approximated by an analytical prescription inspired by studies for fast flavor conversions [17, 18]. We also evaluate the corresponding changes of the ν_e and $\bar{\nu}_e$ heating rates as well as the associated equilibrium electron fraction (Y_e) due to SFC, finding appreciable deviations from their initial (unoscillated) values.

The paper is organized as follows. We introduce in Sec. II the initial neutrino energy spectra considered in this work, and describe the quantum kinetic equations, initial conditions, and numerical setup in Sec. III. Linear stability analysis and nonlinear evolution are presented in Sec. IV and Sec. V, respectively. Sec. VI quantifies the impact of SFC on the physical energy spectra, the resulting heating rates, and the equilibrium Y_e. In Sec. VII, we introduce the phenomenological box prescription and compare it with the simulation outcome. Sec. VIII summarizes our conclusions. In Appendix A, we discuss the numerical convergence and the required simulation resolution. Appendix B includes the complete coarse-grained asymptotic phase-space distributions and compares them with the dominant unstable eigenfunctions. We use natural units with $\hbar = c = 1$ throughout the paper.

II. NEUTRINO ENERGY SPECTRA IN BENCHMARK SCENARIOS

We work in the two-flavor scenario that includes ν_e, ν_x, and their antiparticles, with ν_x denoting the heavy-lepton neutrino, ν_μ or ν_τ, whose properties are taken to be identical. We take three sets of neutrino mean energies and energy luminosities for different species at different post-bounce times, listed in Table I. These values are taken from three snapshots at post-core-bounce time $t_{\rm pb} = 0.05$, 0.2, and 3.0 s from a reference spherically symmetric CCSN simulation for an 18 $M_\odot$ progenitor [19], and represent typical conditions during the early accretion, late accretion, and cooling phases, respectively.

For each species $\nu_\alpha = \nu_e, \bar{\nu}_e, \nu_x, \bar{\nu}_x$, we define the number-density normalization factor relative to the initial ν_e as

$$R_{\nu_\alpha} \equiv \frac{n^{(0)}_{\nu_\alpha}}{n^{(0)}_{\nu_e}} = \frac{L_{\nu_\alpha}/\langle E_{\nu_\alpha}\rangle}{L_{\nu_e}/\langle E_{\nu_e}\rangle}\,, \tag{1}$$

where $n^{(0)}_{\nu_\alpha}$ is the initial number density of ν_α in the considered location. The initially ν_e-normalized, angle-integrated energy spectra for a given species are assumed to follow the parametrization of Ref. [20]:

$$\widehat{f}^{(0)}_{\nu_\alpha}(E_\nu) = R_{\nu_\alpha}\frac{(\chi_{\nu_\alpha}+1)^{\chi_{\nu_\alpha}+1}}{\Gamma(\chi_{\nu_\alpha}+1)}\frac{E_\nu^{\chi_{\nu_\alpha}}}{\langle E_{\nu_\alpha}\rangle^{\chi_{\nu_\alpha}+1}} \times \exp\left[-(\chi_{\nu_\alpha}+1)\frac{E_\nu}{\langle E_{\nu_\alpha}\rangle}\right], \tag{2}$$

where E_ν is the neutrino energy and $\Gamma(x)$ is the Gamma function. Note that $\int_0^\infty dE_\nu\, \widehat{f}^{(0)}_{\nu_\alpha}(E_\nu) = R_{\nu_\alpha}$. We set $\chi_{\nu_\alpha} = 3$ for all species, so their differences are entirely specified by the mean energies and luminosities in Table I.

TABLE I. Benchmark mean energies and energy luminosities for the three CCSN phases taken at different post-bounce time $t_{\rm pb}$ from Ref. [19]. Values in each entry separated by slashes are given in the order of ν_e, $\bar{\nu}_e$, and ν_x, respectively. We assume ν_x and $\bar{\nu}_x$ share identical properties.

Phase	$t_{\rm pb}$ [s]	$\langle E_{\nu_\alpha}\rangle$ [MeV]	L_{ν_α} [10^{51} erg/s]
Early accretion	0.05	8/11/16	36/32/28
Late accretion	0.2	9.6/11.8/16	44/45/20
Cooling	3.0	9.0/10.8/11.5	2.1/2.0/2.5

Panels (a)–(c) of Fig. 1 show the initial spectra for different species, assuming that ν_x and $\bar{\nu}_x$ are identical before flavor oscillations occur. In each phase, the electron- and heavy-flavor spectra cross once in both the neutrino and antineutrino sectors. The ν_e–ν_x and $\bar{\nu}_e$–$\bar{\nu}_x$ crossings occur at $E_\nu \simeq 14.87$ and 17.66 MeV for the early accretion phase spectra, 20.06 and 26.22 MeV for late accretion, and 10.88 and 4.03 MeV for cooling, respectively. At lower energies (below the crossing energy), the electron-flavor spectrum is larger, while the heavy-flavor spectrum dominates above it. For the cooling phase spectra, the low-energy excess of $\bar{\nu}_e$ over $\bar{\nu}_x$ below 4.03 MeV is practically negligible.

A. Lepton-number spectra

For SFC, it is useful to define the lepton-number spectra G_ω as a function of the vacuum oscillation frequency $\omega = \pm|\delta m^2|/(2E_\nu)$, where + (−) corresponds to ν ($\bar{\nu}$) and $|\delta m^2|$ is the absolute value of the neutrino mass-squared difference. For convenience, we work in dimensionless units by setting the representative neutrino-neutrino forward-scattering strength parameter $\mu \equiv \sqrt{2}G_F n^{(0)}_{\nu_e} = 1$, with G_F the Fermi coupling constant, and express all dimensional quantities in units of μ throughout this paper unless specified. We set $\omega = 0.05$ for $E_\nu = 10$ MeV.

Following Eq. (2), the normalized spectrum of each species in the ω space becomes

$$f_{\nu_\alpha}(\omega) = \frac{[(\chi_{\nu_\alpha}+1)\bar{\omega}_{\nu_\alpha}]^{\chi_{\nu_\alpha}+1}}{\Gamma(\chi_{\nu_\alpha}+1)}|\omega|^{-(\chi_{\nu_\alpha}+2)}e^{-(\chi_{\nu_\alpha}+1)\bar{\omega}_{\nu_\alpha}/|\omega|}, \tag{3}$$

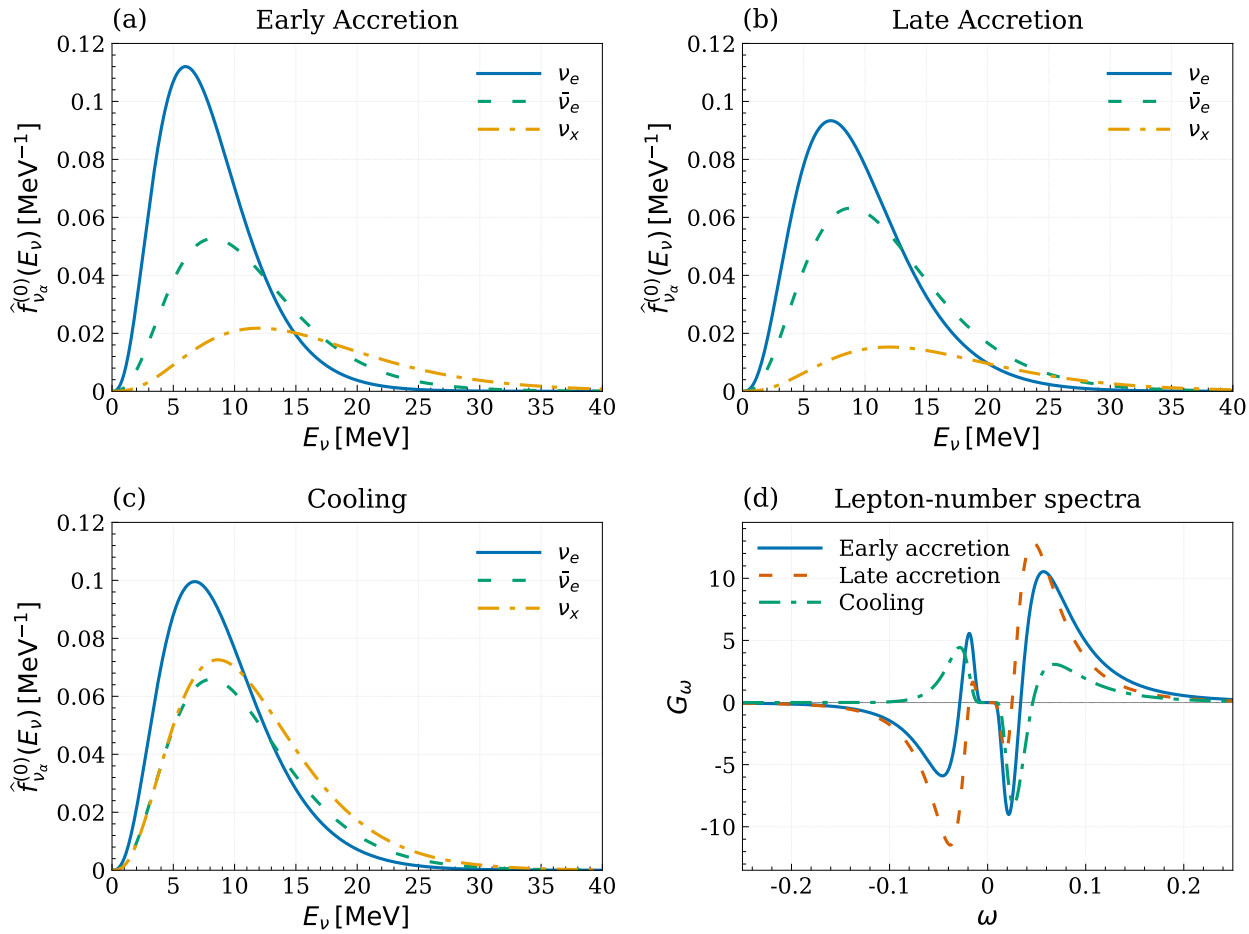


FIG. 1. Initial spectra of the three benchmark phases. Panels (a)–(c) show the ν_e-normalized energy spectra $\widehat{f}^{(0)}_{\nu_\alpha}(E_\nu)$ from Eq. (2) using $\langle E_{\nu_\alpha}\rangle$ and L_{ν_α} values listed in Table I for early accretion, late accretion, and cooling, respectively. Panel (d) shows the corresponding lepton-number spectra G_ω [Eq. (4)]. Note that panel (d) takes the dimensionless convention adopted throughout the text by setting $\mu = \sqrt{2}G_F n^{(0)}_{\nu_e} = 1$ so that ω is in units of μ.

where $\bar{\omega}_{\nu_\alpha} \equiv |\delta m^2|/(2\langle E_{\nu_\alpha}\rangle)$ so that $\int_0^\infty d\omega\, f_{\nu_\alpha}(\omega) = 1$ for neutrinos and $\int_{-\infty}^0 d\omega\, f_{\bar{\nu}_\alpha}(\omega) = 1$ for antineutrinos.

Figure 1(d) shows G_ω defined by

$$G_\omega \equiv \begin{cases} G^+_\omega = f_{\nu_e}(\omega) - R_{\nu_x} f_{\nu_x}(\omega), & \omega > 0, \\ G^-_\omega = R_{\bar{\nu}_x} f_{\bar{\nu}_x}(\omega) - R_{\bar{\nu}_e} f_{\bar{\nu}_e}(\omega), & \omega < 0, \end{cases} \tag{4}$$

for all three benchmark phases. Positive G_ω at $\omega > 0$ denotes a ν_e excess, whereas negative G_ω at $\omega < 0$ denotes a $\bar{\nu}_e$ excess relative to the corresponding heavy flavor. The zeros of G_ω, which correspond to crossings between the electron- and heavy-lepton-flavor energy spectra, are located at $\omega \simeq -0.0283$ and 0.0336 for the early accretion phase, $\omega \simeq -0.0191$ and 0.0249 for late accretion, and $\omega \simeq -0.124$ and 0.0460 for cooling. In all three phases, G_ω also changes sign across $\omega = 0$, where the high-energy tails vanish. While there exist three zero crossings in G_ω for all three phases, the spectrum for the cooling phase essentially represents a two-crossing case, since the antineutrino sector is practically dominated by $\bar{\nu}_x$.

III. QUANTUM KINETIC EQUATIONS AND NUMERICAL SETUP

We adopt the convention that treats both ν and $\bar{\nu}$ on equal footing [4, 21], and use $\rho(\omega > 0)$ and $\rho(\omega < 0)$ to denote their respective reduced density matrices. Assuming translation symmetry in the x and y directions and axisymmetry with respect to the local z direction, the evolution of the dense neutrino gas without momentum-changing collisions is governed by the mean-field quantum kinetic equation (QKE),

$$(\partial_t + v_z\partial_z)\,\rho(t, z, \omega, v_z) \;=\; -i\,[H_{\rm vac} + H_{\nu\nu},\, \rho]\;, \tag{5}$$

where the diagonal and off-diagonal elements of

$$\rho = \begin{pmatrix} \rho_{ee} & \rho_{ex} \\ \rho_{ex}^* & \rho_{xx} \end{pmatrix} , \tag{6}$$

are related to the differential electron-minus-heavy-lepton number (E-XLN) per phase space volume (see the next subsection) and flavor correlations for both ν ($\omega > 0$) and $\bar{\nu}$ ($\omega < 0$), respectively. Note that we assume spatial homogeneity of the matter term originating from the neutrino-electron forward scattering and use an effective in-medium vacuum mixing angle to account for its effect [22].

The vacuum Hamiltonian in the above QKE is given by

$$H_{\rm vac} = \frac{\eta_{\rm MO}\omega}{2} \begin{pmatrix} -\cos 2\theta_V & \sin 2\theta_V \\ \sin 2\theta_V & \cos 2\theta_V \end{pmatrix}, \tag{7}$$

where $\eta_{\rm MO} = +1$ (-1) for normal (inverted) mass ordering and $\theta_V \ll 1$ is the effective in-medium mixing angle. The neutrino self-interaction Hamiltonian is

$$H_{\nu\nu} \;=\; \int_{-\infty}^{\infty} d\omega' \int dv_z' \, (1 - v_z v_z') \rho(\omega', v_z') \; . \tag{8}$$

Since we have set $\mu = \sqrt{2} G_F n_{\nu_e}^{(0)} = 1$, it does not explicitly appear in $H_{\nu\nu}$.

A. Initial conditions and numerical setup

We initialize the reduced density matrices by

$$\rho(t_0, z, \omega, v_z) = \frac{G(\omega, v_z)}{2} \begin{pmatrix} 1 + s(z) & \varepsilon(z) \\ \varepsilon(z) & 1 - s(z) \end{pmatrix}, \tag{9}$$

where

$$G(\omega, v_z) = G_\omega g_\nu(v_z) \tag{10}$$

with $g_\nu(v_z) \propto \exp[-(v_z - 1)^2/(2\sigma_\nu^2)]$, a normalized angular distribution that satisfies $\int dv_z \, g_\nu(v_z) = 1$. Clearly, $G(\omega, v_z)$ represents the ν_e-normalized differential E-XLN spectrum with $\int d\omega dv_z G(\omega, v_z) = 1 - R_{\nu_x} + R_{\bar{\nu}_x} - R_{\bar{\nu}_e}$ after integrating over the entire ω and v_z ranges. We adopt $\sigma_\nu = 0.6$ for every species so that no E-XLN crossing is present in the initial condition, i.e., the flavor conversion studied here is driven by crossings in frequency space. We take $\varepsilon(z) = 10^{-2} \exp\left(-z^2/50\right)$ to seed inhomogeneous and real off-diagonal perturbations. Correspondingly, $s(z) = \sqrt{1 - \varepsilon^2(z)}$ so that the reduced density matrices satisfy $\mathrm{Tr}\, \rho = G$. The seed changes the initial diagonal spectra only at $\mathcal{O}(\varepsilon^2)$, which we neglect when quoting the initial spectra and number densities.

Given the initial conditions, we solve the QKE for each phase until the system has reached a spatially averaged (coarse-grained) quasi-stationary state at about $t \simeq 1000$, using the finite difference version of `COSEν` [23] extended for this work to support the energy dependence explicitly. The simulations use periodic boundary conditions over $z \in [-50, 50]$, a uniform velocity grid over $v_z \in [-1, 1]$, and a uniform frequency grid over $\omega \in [-0.25, 0.25]$. Unless stated otherwise, the results shown below are obtained with $N_z = 1000$, $N_{v_z} = 600$, $N_\omega = 200$, and the Courant–Friedrichs–Lewy (CFL) number of 0.9 for the two accretion phases; for the cooling phase, we use $N_z = 2000$, $N_{v_z} = 800$, and $N_\omega = 200$. The resolution dependence on some of these choices is provided in Appendix A.

Following Ref. [12], we introduce the polarization vector $\mathbf{P}$ of each (ω, v_z) mode, whose components are given by

$$\begin{aligned} P_1 &\equiv +2\,\mathrm{Re}(\rho_{ex})/G \,, \\ P_2 &\equiv -2\,\mathrm{Im}(\rho_{ex})/G \,, \\ P_3 &\equiv (\rho_{ee} - \rho_{xx})/G \,. \end{aligned} \tag{11}$$

Initially, $\mathbf{P} = (\varepsilon, 0, s)$, and unitarity preserves the norm of each polarization vector. Equivalently, the evolved reduced density matrix can be written as

$$\rho = \frac{G}{2} \begin{pmatrix} 1 + P_3 & P_1 - iP_2 \\ P_1 + iP_2 & 1 - P_3 \end{pmatrix} \; . \tag{12}$$

The survival probabilities for an initial ν_e, $\bar{\nu}_e$, ν_x, or $\bar{\nu}_x$ can then be expressed in terms of P_3 as

$$P_{\rm sur}(\omega, v_z) = \frac{1 + P_3(\omega, v_z)}{2} \; . \tag{13}$$

With these definitions, the evolved neutrino distributions at any given time for all species can be written as

$$\begin{aligned} \Phi_{\nu_e}(\omega, v_z) &= R_{\nu_x} f_{\nu_x}(\omega) g_\nu(v_z) + G(\omega, v_z) P_{\rm sur}(\omega, v_z) \,, \\ \Phi_{\nu_x}(\omega, v_z) &= R_{\nu_e} f_{\nu_e}(\omega) g_\nu(v_z) - G(\omega, v_z) P_{\rm sur}(\omega, v_z) \,, \\ \Phi_{\bar{\nu}_e}(\omega, v_z) &= R_{\bar{\nu}_x} f_{\bar{\nu}_x}(|\omega|) g_\nu(v_z) - G(\omega, v_z) P_{\rm sur}(\omega, v_z) \,, \\ \Phi_{\bar{\nu}_x}(\omega, v_z) &= R_{\bar{\nu}_e} f_{\bar{\nu}_e}(|\omega|) g_\nu(v_z) + G(\omega, v_z) P_{\rm sur}(\omega, v_z) \,, \end{aligned} \tag{14}$$

where Φ_{ν_e} and Φ_{ν_x} ($\Phi_{\bar{\nu}_e}$ and $\Phi_{\bar{\nu}_x}$) are defined for $\omega > 0$ ($\omega < 0$).

From Eq. (14), one can compute the time-dependent, spatial- and angular-averaged neutrino energy spectra as

$$\widehat{f}_{\nu_\alpha}(E_\nu, t) = \int dv_z \; \langle \Phi_{\nu_\alpha}(t, z, v_z, \omega(E_\nu)) \rangle_z \left| \frac{d\omega}{dE_\nu} \right| , \tag{15}$$

where $|d\omega/dE_\nu| = |\omega/E_\nu|$ is the Jacobian. One can show that Eq. (15) reduces to Eq. (2) at the initial time.

IV. LINEAR STABILITY ANALYSIS

The linear stability analysis (LSA) helps determine whether a small perturbation in flavor correlation for a

given system configuration will grow exponentially over time (signaling an unstable solution) [24, 25]. It also helps determine the spatial wavenumber at which the instability develops, which constrains the length scales over which flavor evolution takes place. We develop the LSA adapted to our discretized multi-energy, multi-angle system.

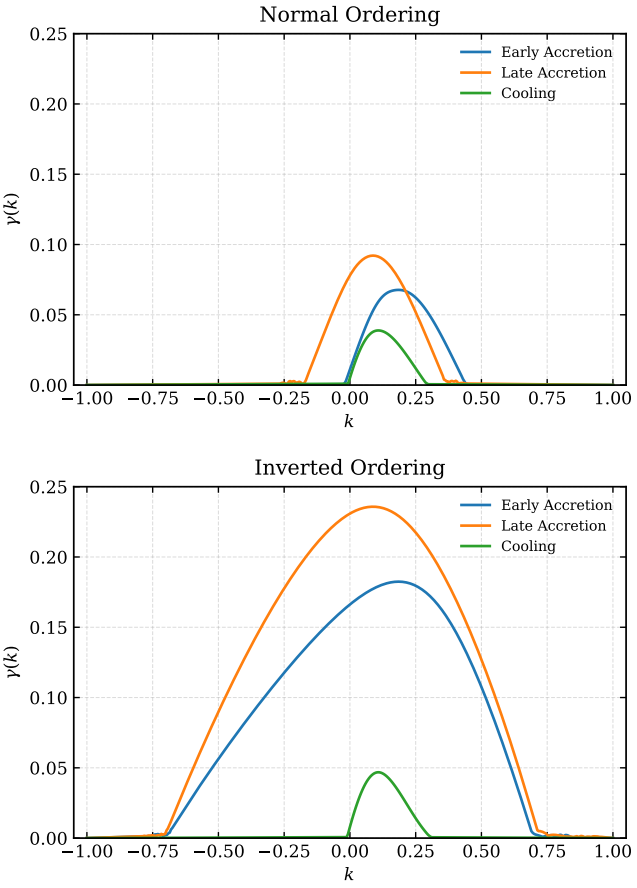


FIG. 2. Unstable growth rate $\mathrm{Im}\,\Omega = \gamma(k)$ for the three benchmark phases in normal (upper) and inverted (lower) mass orderings. Table II reports the maximum growth rate for each phase. The two panels adopt the same y scale for direct comparison. The growth rates in IO are generally larger than in NO. The values of γ and k are in units of μ. [**IP:** The growth rates are unchanged, only the sign convention has been flipped, $k \to -k$. also, i set both panels to the same y axis range so that the NO/IO growth rates are easier to compare]

We linearize the reduced density matrices around the flavor-diagonal state following Ref. [12], writing

$$\rho(t, z, \omega, v_z) = \begin{pmatrix} \rho_{ee}^{(0)} - \Delta & GS/2 \\ GS^*/2 & \rho_{xx}^{(0)} + \Delta \end{pmatrix}. \tag{16}$$

Here, $\rho_{\alpha\alpha}^{(0)}$ denotes the unperturbed reduced spectra obtained by taking $\varepsilon \to 0$ in the initial conditions above. Thus, $\rho_{ee}^{(0)} = G$, $\rho_{xx}^{(0)} = 0$ for both neutrinos and antineutrinos. The perturbation is represented by $S = P_1 - iP_2 = 2\rho_{ex}/G$. Note that unitarity gives $\Delta = G[1 - \sqrt{1 - |S|^2}]/2 = \mathcal{O}(|S|^2)$ and therefore the diagonal evolution is subleading in the linear regime. As a result, only the equation for S is retained.

TABLE II. Dominant growth rate $\gamma_{\max}$ extracted from Eq. (19), and the wavenumber $k_\star$ at which it develops, for the three CCSN phases. The values of $\gamma_{\max}$ and k_* are in units of μ.

Phase	$\gamma_{\max}$ (NO/IO)	$k_\star$
Early accretion	0.066/0.18	0.18
Late accretion	0.091/0.23	0.08
Cooling	0.039/0.047	0.10

Substituting the commonly used plane-wave ansatz

$$S(t, z, \omega, v_z) = Q(\omega, v_z)\, e^{-i(\Omega t - kz)}, \tag{17}$$

into Eq. (5) and keeping only terms linear in Q, one obtains the following eigenvalue equation

$$\Omega\, \vec{Q} = M(k)\, \vec{Q}\ , \tag{18}$$

where $\vec{Q}$ has components $Q_{ij} \equiv Q(\omega_i, v_j)$ on the discrete (ω, v_z) grid. The dominant (largest) growth rate at wavenumber k is

$$\gamma(k) \equiv \max_n \mathrm{Im}\,\Omega_n(k)\ , \tag{19}$$

where $\Omega_n(k)$ is the nth eigenvalue of the matrix $M(k)$ and n labels the discrete eigenmodes. The overall fastest-growing mode of the system is $\gamma_{\max} \equiv \max_k \gamma(k)$, which corresponds to the maximum value among all k modes. The matrix elements of M read

$$\begin{aligned} M_{(i,a)(j,b)} = \delta_{ij}\delta_{ab} \big[&- \eta_{\rm MO}\omega_i + k v_a + \eta(1 - v_a \langle v \rangle_g)\big] \\ &- G_{\omega_j} g_\nu(v_b)(1 - v_a v_b)\, \Delta\omega\, \Delta v\ , \end{aligned} \tag{20}$$

where the (i, a) and (j, b) indices label the (ω, v_z) pairs, and

$$\eta \equiv \int d\omega\, G_\omega\ , \qquad \langle v \rangle_g \equiv \int dv_z\, v_z\, g_\nu(v_z)\ . \tag{21}$$

These are the integrated lepton-number asymmetry and the velocity-weighted mean of the angular distribution. To avoid clutter in Eqs. (20), we have dropped the subscript z on the velocity variables.

For each supernova phase of Table I, we construct $M(k)$ on a uniform velocity grid in $v_z \in [-1, 1]$ with $N_v = 100$ and a uniform grid of $N_\omega = 80$ over $\omega \in [-0.25, 0.25]$. The matrix is diagonalized with a standard dense eigenvalue solver from `numpy` [26] at each of $N_k = 199$ wavenumbers uniformly distributed in $k \in [-1, 1]$, and the dominant growth rate $\mathrm{Im}\,\Omega$ is kept for each k. The dispersion relation $\gamma(k)$ identifies a band of unstable wavenumbers in each phase, with the maximum growth rate $\gamma_{\max}$ tabulated in Table II.

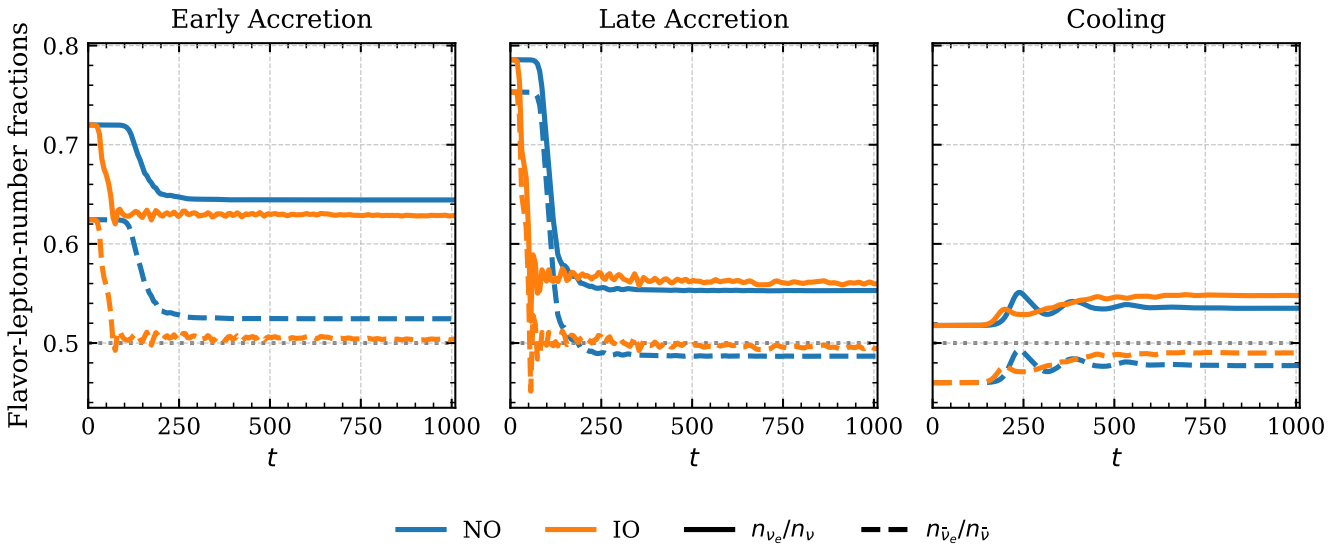


FIG. 3. Time evolution of the flavor-lepton-number fractions n_{ν_e}/n_ν (solid) and $n_{\bar{\nu}_e}/n_{\bar{\nu}}$ (dashed) for the early accretion (left panel), late accretion (middle panel), and cooling (right panel) phases, in NO (blue) and IO (orange), with t in units of μ^{-1}. Here $n_\nu \equiv n_{\nu_e} + n_{\nu_x}$ and $n_{\bar{\nu}} \equiv n_{\bar{\nu}_e} + n_{\bar{\nu}_x}$, both energy- and angle-integrated and averaged over the box. The horizontal dotted line at 1/2 marks flavor equipartition.

Figure 2 shows the dispersion relations for all three phases in both mass orderings. For the cooling phase in NO, the instability is confined to a narrow band of positive wavenumbers around $k \simeq 0.1$ and reaches a peak growth rate of $\gamma_{\rm max} \simeq 0.039$; modes outside this band have growth rates compatible with zero within numerical noise from spurious modes. The two accretion phases behave differently in the two orderings: in NO their unstable bands are only slightly broader than the cooling one, with $\gamma_{\rm max} \simeq 0.066$ and 0.091 for early and late accretion, respectively, whereas in IO both bands extend over $|k| \lesssim 0.7$ and the peak growth rates increase to 0.18 and 0.23. The cooling phase, by contrast, is almost insensitive to the ordering (0.039 versus 0.047). This insensitivity of the cooling phase to the mass ordering can be traced to the sign structure of G_ω. For the cooling spectrum, the integrated neutrino and antineutrino sectors are both positive, $\int_{\omega>0} d\omega\, G_\omega \simeq 0.068$ and $\int_{\omega<0} d\omega\, G_\omega \simeq 0.138$, because the antineutrino sector is almost entirely dominated by $\bar{\nu}_x$. By contrast, the corresponding integrals have opposite signs for early accretion, $(0.611, -0.258)$, and late accretion, $(0.727, -0.559)$. An explicit decomposition of the $k' = 0$ mode (see Appendix C), demonstrates that the cooling spectrum excites the same, strongly coupled angular eigenmode in *both* mass orderings, whereas the accretion spectra excite the weaker angular eigenmode in NO and the stronger one in IO. This feature accounts for the much weaker ordering dependence of the cooling phase growth rate and is in fact robust under variations of the neutrino angular distributions, as shown in Appendix C.

Moreover, $k_\star$ can be cross-checked from the initial background (E-X)LN current,

$$J_z \equiv \int d\omega\, dv_z\, v_z\, G(\omega, v_z) = \eta \langle v \rangle_g \ . \tag{22}$$

Notice that the velocity-dependent diagonal terms in Eq. (20) combine as $v_a(k - J_z)$. For all three phases and both mass orderings, the fastest growing mode is the zero mode of the shifted wavenumber, $k' = k - J_z = 0$, and therefore $k_\star = J_z$, independently of the mass ordering. The shifted wavenumber absorbs the phase induced by the current J_z, following the dispersion relation formulation of Ref. [25]. When $k' = 0$, the diagonal term proportional to $v_a k'$ vanishes in Eq. (20), eliminating this source of kinematic phase dispersion among angular modes. Numerically, this gives $k_\star = (0.185, 0.088, 0.108)$ for the early accretion, late accretion, and cooling phases, respectively, in agreement with Table II.

The LSA growth rates estimated here (Fig. 2) agree well with those extracted from the early exponential growth of the nonlinear solutions presented in the next section for all three supernova phases.

V. NON-LINEAR FLAVOR EVOLUTION

We show in Fig. 3 the evolution of the coarse-grained, energy- and angle-integrated fractions n_{ν_e}/n_ν and $n_{\bar{\nu}_e}/n_{\bar{\nu}}$ for all three phases and both mass orderings, where $n_\nu \equiv n_{\nu_e} + n_{\nu_x}$ and $n_{\bar{\nu}} \equiv n_{\bar{\nu}_e} + n_{\bar{\nu}_x}$. In every case, the ratios stay near their initial values during the linear phase, depart rapidly once the instability saturates, and then evolve toward quasi-stationary values by $t \simeq 1000$.

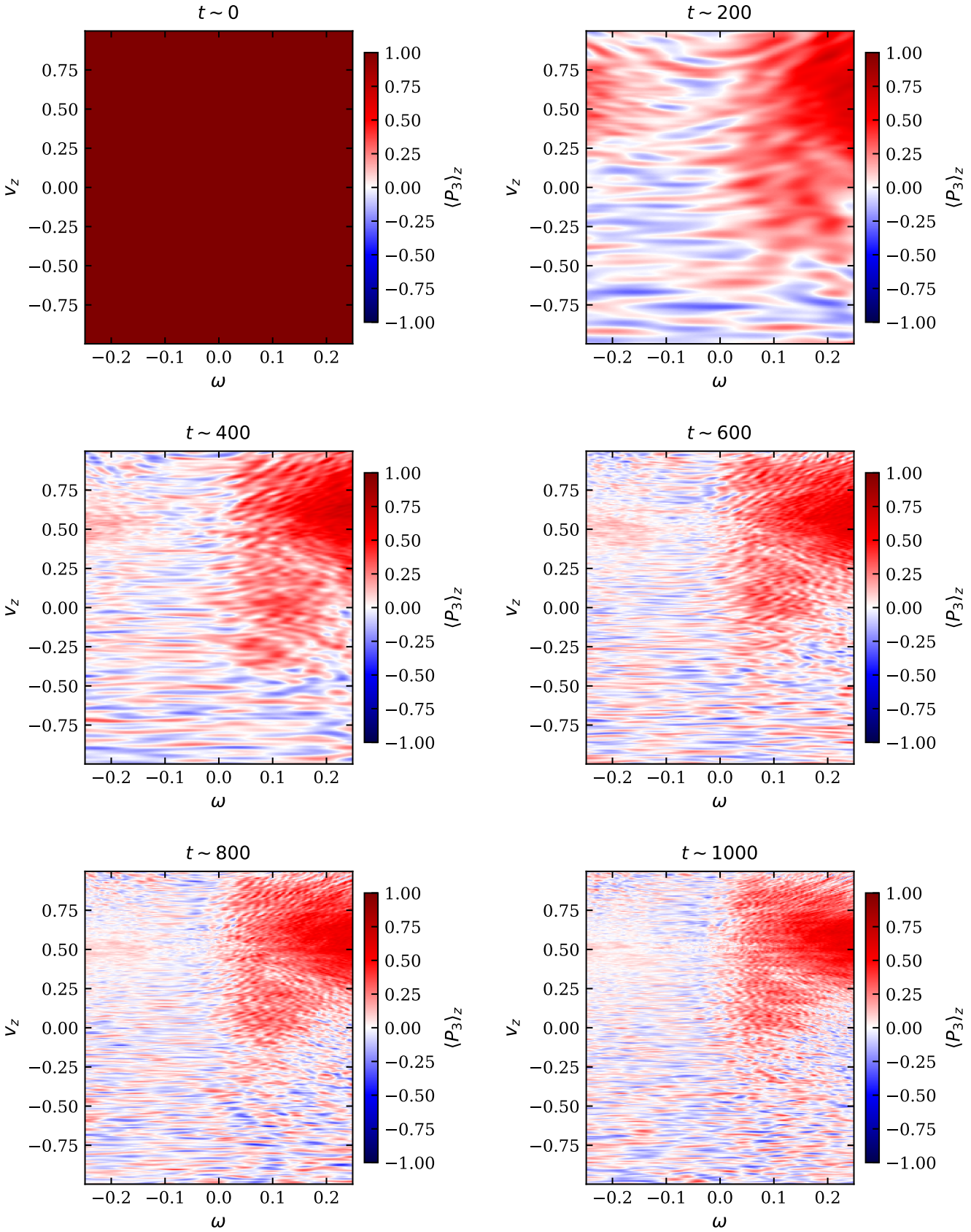


FIG. 4. Snapshots of the spatially averaged flavor polarization $\langle P_3 \rangle_z$ in the (ω, v_z) plane at six times, for the late accretion phase in IO. The quantities t and ω are in units of μ^{-1} and μ, respectively.

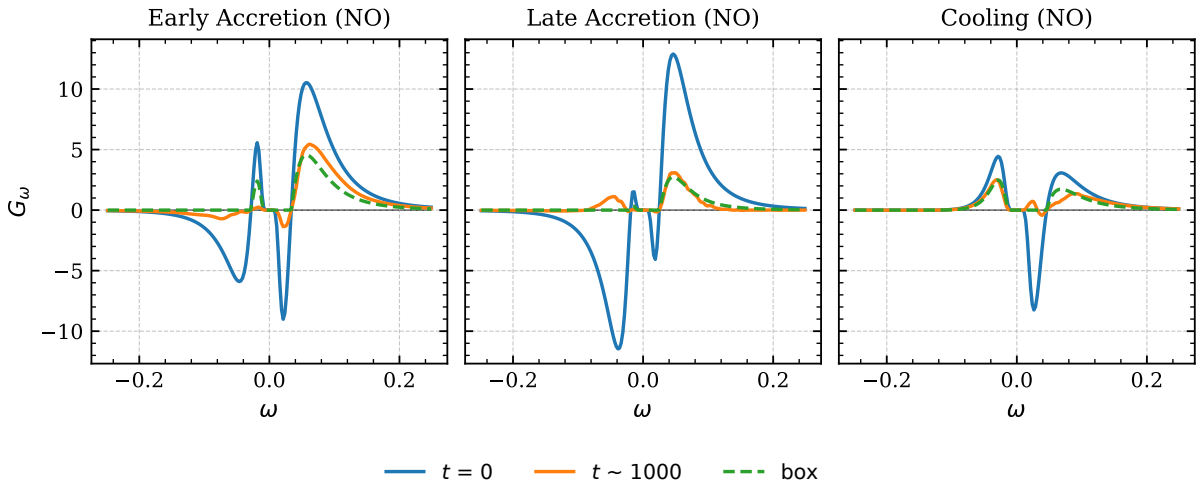


FIG. 5. Comparison of the initial ($t = 0$, blue) and asymptotic ($t \simeq 1000$, orange) lepton-number spectra G_ω, averaged over z and integrated over v_z, for the early accretion (left panel), late accretion (middle panel), and cooling (right panel) phases in NO. Also shown are the final G_ω computed with the box recipe (green dashed curve); see Sec. VII for details. The quantities t and ω are in units of μ^{-1} and μ, respectively.

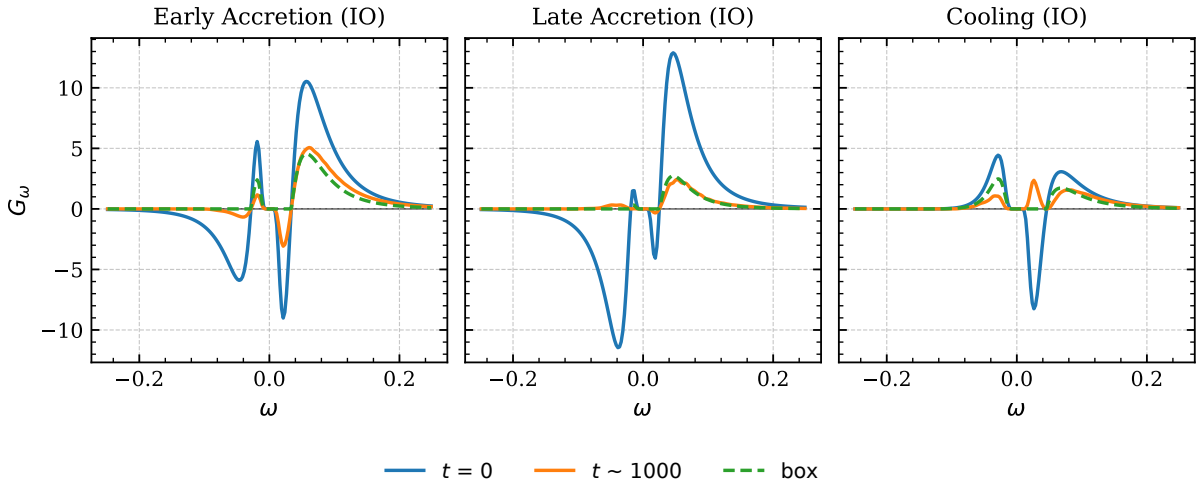


FIG. 6. Same as Fig. 5, but for IO.

The saturation time tracks the growth rates of Table II: the late accretion phase, which has the largest $\gamma_{\max}$, departs from its initial state first, while the cooling phase is the slowest.

For the early and late accretion phases, both n_{ν_e}/n_ν and $n_{\bar{\nu}_e}/n_{\bar{\nu}}$ decrease from their initial values to lower final values, reflecting the net conversion of electron flavor to heavy-lepton flavor because $R_{\nu_e}/R_{\nu_x} > 1$ and $R_{\bar{\nu}_e}/R_{\bar{\nu}_x} > 1$. In contrast, $R_{\nu_e}/R_{\nu_x} < 1$ and $R_{\bar{\nu}_e}/R_{\bar{\nu}_x} < 1$ for the cooling phase, so that SFC results in a net increase of n_{ν_e}/n_ν and $n_{\bar{\nu}_e}/n_{\bar{\nu}}$. For all phases, the different mass orderings only moderately affect the final values by $\lesssim 0.03$, consistent with the findings of Ref. [12]. For all cases, $n_{\bar{\nu}_e}/n_{\bar{\nu}}$ in IO appears to reach a value closer to 0.5, i.e., equipartition, than in NO, which is also consistent with Ref. [12]. For $n_{\bar{\nu}_e}/n_{\bar{\nu}}$, they asymptote to final values constrained by approximate E-XLN conservation [12].

Fig. 4 further shows the spatially averaged polarization vector component $\langle P_3 \rangle_z$ in the (ω, v_z) plane at six times for the late accretion phase in IO as an example. First, $\langle P_3 \rangle_z$ develops a pronounced dependence on both energy and angle. The neutrino modes with $\omega \gtrsim \omega_c \simeq 0.0249$ and with $v_z \gtrsim 0$ retain $\langle P_3 \rangle_z \simeq 0.3$–$0.8$ and therefore undergo little flavor conversion. In contrast, the entire antineutrino half-axis, the high-energy neutrinos with $\omega \lesssim \omega_c$, and the backward-moving modes with $v_z \lesssim 0$ all relax to $\langle P_3 \rangle_z \simeq 0$, i.e., near flavor equipartition. Second, both the angular and the energy structures become progressively finer with time. In particular, the cascade in v_z is stronger, producing finer horizontal stripes; see

Appendix A for further discussion.

In Figs. 5 and 6, we compare the spatially averaged and angle-integrated initial and asymptotic lepton-number spectra G_ω of the three phases in NO and IO, respectively. These plots show that the final-state G_ω can generally still contain spectral crossings, clearly exemplified by the early accretion phase in both NO and IO, although the depths of the negative G_ω are substantially reduced by SFC. We speculate that this may be consistent with the conclusion of Ref. [27] that the presence of multiple weak spectral crossings may not guarantee a flavor instability. For the late accretion and cooling phases, slight flavor overconversion generally occurs for the lobes whose G_ω are negative initially, consistent with the result of Ref. [12] for single-energy cases. Whether this overconversion is related to the initially imposed strong flavor instability associated with deep spectral crossings, see, e.g., Ref. [28, 29] for fast flavor conversions, which may not actually develop in realistic astrophysical environments, remains to be clarified by future work.

Among all cases, nearly perfect flavor equipartition only occurs for the two middle lobes associated with small $|\omega|$ (high-energy ν and $\bar{\nu}$ above the crossing energy) in the late accretion phase, although for all cases the high-energy neutrino modes (small ω) reach a final state relatively close to equipartition. In contrast, the largest positive lobe in the neutrino sector (large ω) experiences the least flavor conversion to preserve the approximate E-XLN conservation condition. Also noted is that the positive G_ω lobe in $\omega < 0$ (antineutrinos) in both cooling phase cases remains largely positive instead of reaching flavor equipartition[2]. Overall, these results clearly show that the extent of flavor conversions due to SFC is energy dependent. As a consequence, the impact of SFC on physical quantities that are more sensitive to the neutrino spectrum at high energy, such as heating and reaction rates, cannot be well captured by treatments adopting the single-energy assumption.

VI. IMPACT ON ENERGY SPECTRA, HEATING RATES AND $Y_e^{\rm eq}$

The energy-dependent modification of the neutrino energy spectra discussed in the previous section has several physical consequences. In addition to the change of the (anti)neutrino spectra, relevant to the neutrino signals expected from the next galactic supernova, it also leads to the modification of the charged-current absorption rates on free nucleons in the post-shocked region as well as in the neutrino-driven wind. The spectral change affects both the heating rate behind the supernova shock and the equilibrium electron fraction, and hence potentially impacts the supernova explosion dynamics and the associated wind nucleosynthesis. In this section, we compute the coarse-grained, angle-integrated, post-SFC energy spectra for all cases that we have examined, and use them to evaluate the corresponding changes to the neutrino heating rates as well as the neutrino absorption equilibrium Y_e. We note that these evaluations are meant to provide clues about how SFC may affect relevant quantities for CCSN physics within the considered scenarios. The actual impact on CCSN dynamics and nucleosynthesis requires a self-consistent implementation of SFC into CCSN simulations, which is beyond the scope of this work.

A. Energy spectra

We use Eq. (15) to reconstruct the physical energy spectra of different species and show the results in Figs. 7 and 8 for all three phases in NO and IO, respectively, where the physical energy units have been restored. As indicated previously for G_ω, above the crossing energy in the neutrino sector, the final ν_e and ν_x spectra become much more similar, approaching flavor equipartition for both NO and IO. In contrast, below the crossing energy the conversion remains partial due to the E-XLN conservation. For antineutrinos, the post-SFC energy spectra are rather close to full equipartition over the entire energy range in both accretion phases, whereas the cooling-phase spectra differ substantially from equipartition for both NO and IO, consistent with Figs. 5 and 6. We note that the energy-dependent imprint of SFC on the neutrino energy spectra is distinct from the effect of FFC; see, e.g., Fig. 21 of Ref. [30]. As a result, detailed extraction of CCSN neutrino energy spectra of different species, if achievable, can potentially be used as diagnostics for the dominant mode of collective flavor conversions.

B. Heating Rates

The neutrino heating rates due to the charged-current absorption of ν_e and $\bar{\nu}_e$ on free nucleons are given by [31]

$$\dot{q}_{\nu_e n} \propto \int_0^\infty [E_\nu + (\Delta - m_e)](E_\nu + \Delta)^2 \Big(1 - \frac{W_{\nu_e} E_\nu}{m_N}\Big) \widehat{f}_{\nu_e}(E_\nu)\, dE_\nu, \tag{23}$$

and

$$\dot{q}_{\bar{\nu}_e p} \propto \int_{E_{th}}^\infty [E_\nu - (\Delta - m_e)](E_\nu - \Delta)^2 \Big(1 - \frac{W_{\bar{\nu}_e} E_\nu}{m_N}\Big) \widehat{f}_{\bar{\nu}_e}(E_\nu) dE_\nu, \tag{24}$$

where $\Delta = m_n - m_p$ is the neutron–proton mass difference, $E_{th} = \Delta + m_e$ is the absorption energy threshold, and

$$W_{\nu_e(\bar{\nu}_e)} = \frac{2[1 + 5g_A^2 - (+)2g_A(1 + f_2)]}{1 + 3g_A^2} \tag{25}$$

is the weak-magnetism correction [32] with $g_A = 1.27$ and $f_2 = 3.71$. To evaluate the change of these heating

[2] This is in contradiction with Ref. [13], which claims flavor equipartition for antineutrinos.

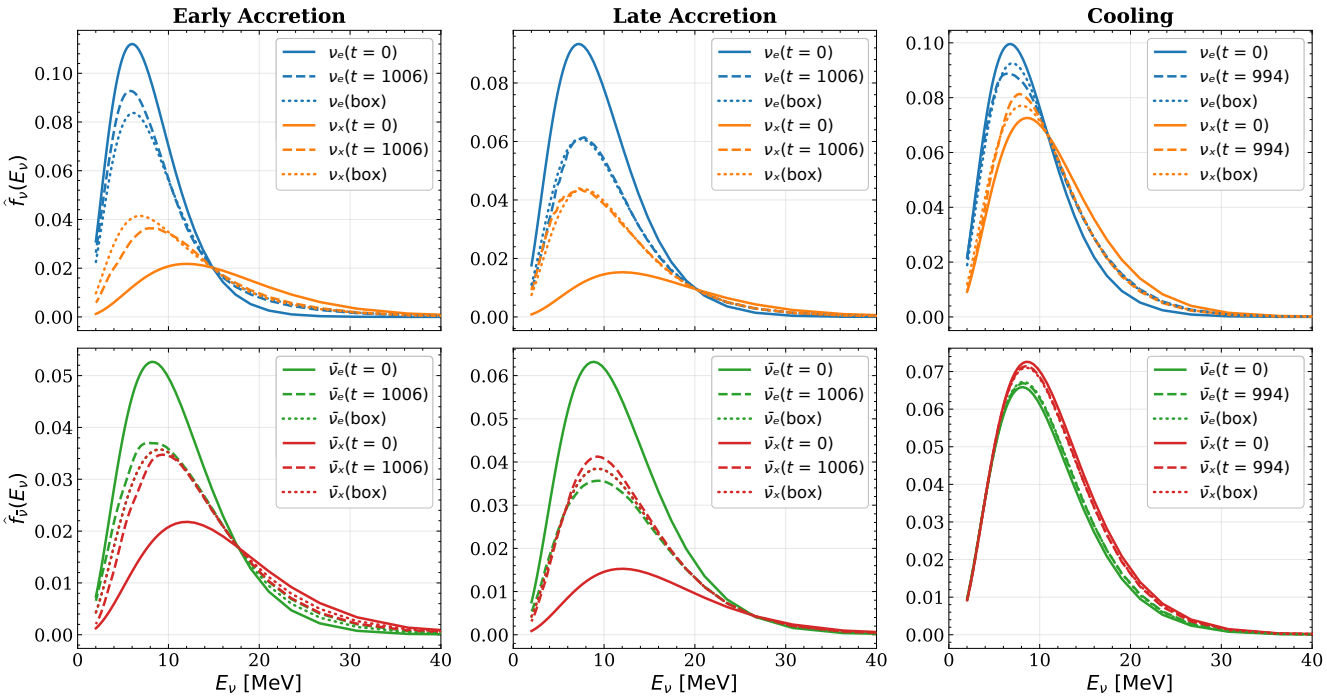


FIG. 7. Comparison of the initial ($t = 0$, solid) and asymptotic ($t \simeq 1000$, dashed) energy spectra $\widehat{f}_{\nu_\alpha}(E_\nu)$, reconstructed with Eq. (15), for the early accretion (left panel), late accretion (middle panel), and cooling (right panel) phases in NO. The upper row shows ν_e (blue) and ν_x (orange), and the lower row shows $\bar{\nu}_e$ (green) and $\bar{\nu}_x$ (red). The spectra are averaged over space and integrated over velocity. The color-matched dotted curves show the spectra computed using the box recipe (see Sec. VII). t is in units of μ^{-1}.

TABLE III. Ratios of the final (post-SFC) to initial heating rates $\dot{q}^f/\dot{q}^i$ for the three CCSN phases in normal and inverted mass orderings. The values of the ratios for different mass orderings are separated by slashes.

Phase	$\dot{q}^f_{\nu_e n}/\dot{q}^i_{\nu_e n}$ (NO/IO)	$\dot{q}^f_{\bar{\nu}_e p}/\dot{q}^i_{\bar{\nu}_e p}$ (NO/IO)
Early accretion	1.84 / 1.62	1.39 / 1.31
Late accretion	1.12 / 1.10	0.89 / 0.90
Cooling	1.47 / 1.53	1.12 / 1.18

rates due to SFC, we compute them using the initial and final spectra shown in Figs. 7 and 8 and denoted the corresponding rates by $\dot{q}^i$ and $\dot{q}^f$, respectively.

Table III shows $\dot{q}^f_{\nu_e n}/\dot{q}^i_{\nu_e n}$ and $\dot{q}^f_{\bar{\nu}_e p}/\dot{q}^i_{\bar{\nu}_e p}$ for all three phases and both mass orderings. The ν_e heating rate is enhanced in all phases, with the largest increase during early accretion: 84% in NO and 62% in IO. This enhancement is driven by near flavor equipartition at high energies above the crossing energy, which raises the high-energy tail of the post-conversion $\widehat{f}_{\nu_e}$. The $\bar{\nu}_e$ heating rate is enhanced by 39% (31%) during early accretion and by 12% (18%) during cooling in NO (IO). For late accretion, it instead decreases by 11% in NO and 10% in IO, because the high-energy tails of the $\bar{\nu}_e$ and $\bar{\nu}_x$ spectra are too similar. Consequently, the reduction of the lower-energy part of the $\bar{\nu}_e$ spectrum results in a net decrease of the $\bar{\nu}_e$ heating rate.

C. Absorption equilibrium electron fraction

The neutrino absorption equilibrium electron fraction, which is a key quantity for nucleosynthesis conditions in neutrino-driven winds [33], is defined by

$$Y_e^{\rm eq} = \left(1 + \frac{\lambda_{\bar{\nu}_e p}}{\lambda_{\nu_e n}}\right)^{-1} , \tag{26}$$

with

$$\lambda_{\nu_e n} \propto \int_0^\infty (E_\nu + \Delta)^2 \left(1 - \frac{W_{\nu_e} E_\nu}{m_N}\right) \widehat{f}_{\nu_e}(E_\nu)\, dE_\nu , \tag{27}$$

$$\lambda_{\bar{\nu}_e p} \propto \int_{E_{th}}^\infty (E_\nu - \Delta)^2 \left(1 - \frac{W_{\bar{\nu}_e} E_\nu}{m_N}\right) \widehat{f}_{\bar{\nu}_e}(E_\nu)\, dE_\nu , \tag{28}$$

denoting the neutrino absorption rates on nucleons. Table IV lists its value computed using the initial and final spectra for all three phases.

Here, we see that SFC leads to an increase in $Y_e^{\rm eq}$ for both NO and IO in all phases by 0.017–0.036. This is due

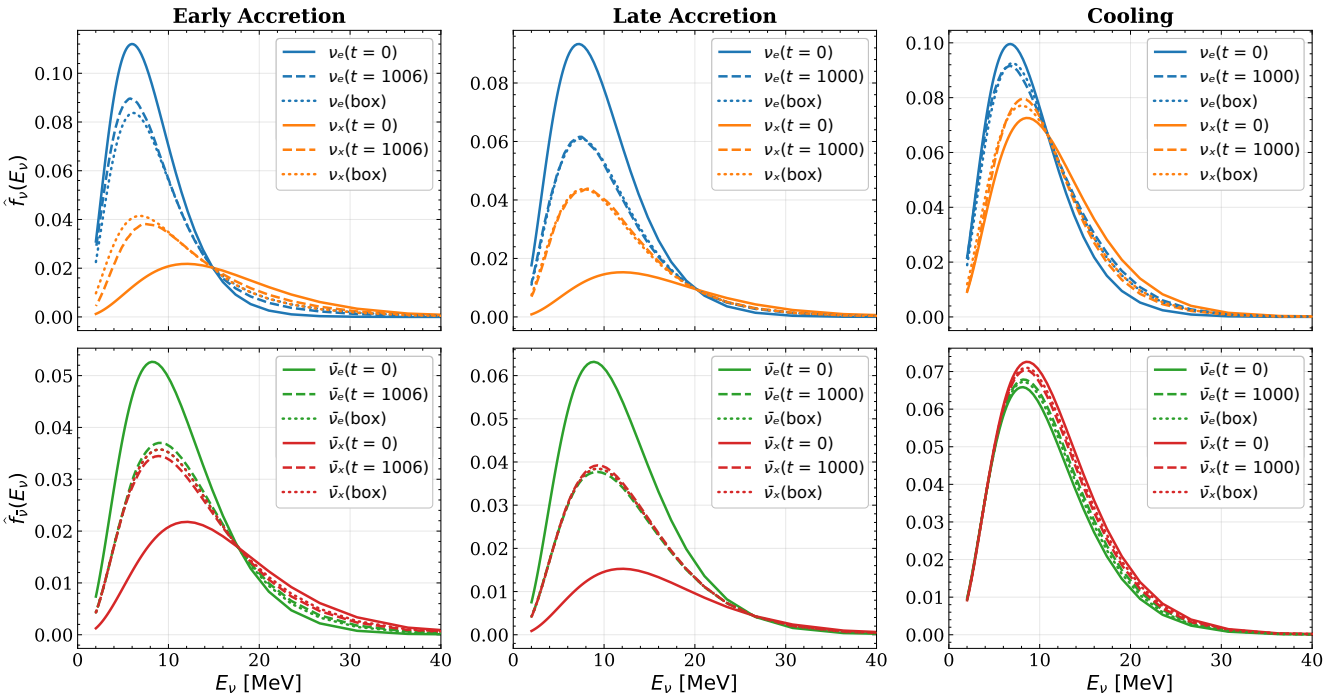


FIG. 8. Same as Fig. 7, but for IO.

TABLE IV. Initial and final (post-SFC) equilibrium electron fractions for the three CCSN phases.

Phase	$Y_e^{\rm eq,i}$	$Y_e^{\rm eq,f}$ (NO)	$Y_e^{\rm eq,f}$ (IO)
Early accretion	0.597	0.624	0.614
Late accretion	0.580	0.615	0.609
Cooling	0.607	0.643	0.641

to the reduced value of $\lambda_{\bar{\nu}_e p}/\lambda_{\nu_e n}$ as $\lambda_{\nu_e n}$ is enhanced more than $\lambda_{\bar{\nu}_e p}$. As a result, SFC can lead to more proton-rich conditions in the ν-driven outflows and potentially favor the production of proton-rich nuclei, particularly if the conditions for the νp-process [34–37] can be realized in the outflow.

VII. A PHENOMENOLOGICAL RECIPE FOR QUASISTATIONARY ENERGY SPECTRA

An important question is whether the coarse-grained quasistationary state due to SFC can be predicted by knowing the initial spectrum of $G(\omega, v_z)$ without performing a full nonlinear QKE calculation. While we find that the final $\langle P_3(\omega, v_z)\rangle_z$ shows a rather complicated structure, which may be related to the shape of the eigenfunctions of the initial unstable modes for some cases (see Appendix B), it appears challenging to find a simple analytical fit to approximate the full $G(\omega, v_z)$ satisfactorily. In this section, we instead investigate whether the coarse-grained, angle-averaged spectra can be well approximated by simple analytical prescriptions.

Assuming that SFC tends to erase (or largely reduce) the initial spectral crossings as shown in earlier sections, we attempt to approximate the final shape of G_ω by imposing a box-like, ω-dependent flavor survival probability, motivated by fast flavor conversion studies [17, 18][3]. We define I_- and I_+ as the absolute area of the negative and positive portions of the lepton-number spectrum, respectively, so that

$$I_- \equiv \left| \int_{G_\omega<0} d\omega\, G_\omega \right| , \qquad I_+ \equiv \int_{G_\omega>0} d\omega\, G_\omega \ . \tag{29}$$

We then formulate the asymptotic survival probability by

$$P^{\rm box}(\omega) = \begin{cases} p_{\rm eq} \ , & G_\omega > 0 \ , \\ 1-(1-p_{\rm eq})\,\dfrac{I_+}{I_-} \ , & G_\omega < 0 \ , \end{cases} \qquad (I_- \geq I_+) \ , \tag{30}$$

and

$$P^{\rm box}(\omega) = \begin{cases} 1-(1-p_{\rm eq})\,\dfrac{I_-}{I_+} \ , & G_\omega > 0 \ , \\ p_{\rm eq} \ , & G_\omega < 0 \ , \end{cases} \qquad (I_+ > I_-) \ , \tag{31}$$

[3] This recipe differs from that proposed in Ref. [13] for cases without E-XLN angular crossings, where full flavor equipartition for antineutrinos is assumed.

TABLE V. Ratio of the box-estimated to simulated heating rate, $\dot{q}_{\rm est}/\dot{q}_{\rm sim}$. Each entry lists NO/IO, and values close to 1 indicate good agreement.

Phase	ν_e (NO/IO)	$\bar{\nu}_e$ (NO/IO)
Early accretion	1.067/1.214	0.859/0.910
Late accretion	1.009/1.031	0.988/0.984
Cooling	0.979/0.942	0.975/0.928

where $p_{\rm eq} = 1/2$ enforces flavor equipartition for $G_\omega > 0$ ($G_\omega < 0$) when $I_- \geq I_+$ ($I_- < I_+$). The associated spectrum is then

$$G_\omega^{\rm box} = \left(2P^{\rm box}(\omega) - 1\right) G_\omega^{\rm initial} \ . \tag{32}$$

We show $G_\omega^{\rm box}$ as well as the corresponding $\widehat{f}_\nu(E_\nu)$ together with the nonlinear results in Figs. 5–8. Since the prescription depends only on the initial G_ω, the box prediction is identical for NO and IO. Figs. 5 and 6 show that the resulting spectrum qualitatively agrees with the simulation outcome, although it fails to capture the flavor overconversion in the late accretion and cooling phases, as well as the residual negative G_ω in the early accretion phase. Similarly, the predicted energy spectra displayed in Figs. 7 and 8 show generally good agreement with the simulation results.

To quantify the performance of the above box recipe, we define $\dot{q}_{\rm est}/\dot{q}_{\rm sim}$ as the ratio of the heating rate obtained from the box-generated spectrum to that obtained from the simulation result, and report the values for all cases in Table V. The table shows that the box recipe accurately reproduces the changes in the heating rates within 10% for most cases. The only exceptions are $\dot{q}_{\nu_e}$ for IO and $\dot{q}_{\bar{\nu}_e}$ for NO in the early accretion phase, where $G_\omega^{\rm box}$ deviates the most from the simulated G_ω at small $|\omega|$ (see Figs. 5 and 6).

In addition, we further quantify the agreement of the energy spectra in Figs. 7 and 8 by defining the simulation-normalized net percentage differences

$$\begin{aligned}
\Delta_\alpha^{\rm box} &\equiv 100 \times \frac{\displaystyle\int dE_\nu \left|\widehat{f}_\alpha^{\rm box}(E_\nu) - \widehat{f}_\alpha^{\rm sim}(E_\nu)\right|}{\displaystyle\int dE_\nu\, \widehat{f}_\alpha^{\rm sim}(E_\nu)}, \\
\Delta_{\rm all}^{\rm box} &\equiv 100 \times \frac{\displaystyle\sum_\alpha \int dE_\nu \left|\widehat{f}_\alpha^{\rm box}(E_\nu) - \widehat{f}_\alpha^{\rm sim}(E_\nu)\right|}{\displaystyle\sum_\alpha \int dE_\nu\, \widehat{f}_\alpha^{\rm sim}(E_\nu)},
\end{aligned} \tag{33}$$

where the summation in the second equation runs over all species.

Table VI reports $\Delta_\alpha^{\rm box}$ and $\Delta_{\rm all}^{\rm box}$ for all cases. Consistently, the early accretion phase shows the largest deviation, particularly for ν_x. Otherwise, the agreement is generally within 5–10%.

TABLE VI. Simulation-normalized percentage differences between the box-predicted and simulated energy spectra [Eq. (33)]. The paired columns list the electron/heavy-flavor values.

Phase	MO	$\Delta_{\nu_e}^{\rm box}/\Delta_{\nu_x}^{\rm box}$	$\Delta_{\bar{\nu}_e}^{\rm box}/\Delta_{\bar{\nu}_x}^{\rm box}$	$\Delta_{\rm all}^{\rm box}$
Early accretion	NO	7.36/13.20	7.02/7.74	8.55
	IO	7.21/12.13	3.35/3.41	6.61
Late accretion	NO	3.06/3.79	4.17/3.95	3.70
	IO	1.95/2.47	1.26/1.23	1.74
Cooling	NO	2.93/3.32	0.80/0.73	2.00
	IO	3.61/4.29	2.61/2.50	3.28

VIII. DISCUSSION AND CONCLUSIONS

In this work, we have extended our previous single-energy study of SFC to a multi-energy, multi-angle system for three neutrino spectra representing different CCSN phases. The linear stability analysis and nonlinear QKE calculations show that SFC is predominantly triggered by spatially inhomogeneous instabilities. For the two accretion phase cases, the maximum growth rates are substantially larger in IO than in NO, while the ordering dependence is weaker for the cooling phase case.

We find that the resulting coarse-grained, quasistationary flavor conversion probability depends on energy, angle, and the neutrino mass ordering. In contrast to the single-energy system, the initial spectral crossings are not completely erased in the coarse-grained, angle-averaged final state. High-energy modes at small $|\omega|$ generally undergo stronger flavor conversion, while lower-energy modes retain a larger fraction of their initial flavor, independent of the mass ordering. The final spectra can exhibit residual crossings as well as flavor overconversion in some cases. These features demonstrate that the post-SFC spectra cannot be represented by a simplified monochromatic treatment.

We have also investigated the impact of SFC on the charged-current neutrino interaction rates. Across all cases considered in this work, the ν_e absorption heating rates are enhanced by 10%–84%. For $\bar{\nu}_e$, the heating rate exhibits a weaker enhancement or even a slight suppression, since its energy spectrum is closer to that of $\bar{\nu}_x$. For both mass orderings and all three phases, SFC increases $Y_e^{\rm eq}$ by 0.017–0.036. This indicates that SFC can potentially affect neutrino heating as well as result in more proton-rich conditions relevant to the yields of proton-rich nuclei in the neutrino-driven wind, if the spectral changes observed in this work are representative.

We have also tested a phenomenological box prescription that estimates the asymptotic coarse-grained and angle-averaged energy spectra directly from the initial spectra. Although the prescription is independent of the mass ordering and cannot reproduce the flavor overconversion and residual crossings obtained in simulations, it reasonably approximates the simulated energy spectra,

with relative spectral differences for individual species and heating-rate deviations generally $\lesssim \mathcal{O}(5\text{–}20\%)$.

While our work has improved the understanding of the evolution and the local coarse-grained quasistationary state of SFC for supernova neutrinos, several aspects remain to be further investigated. The formal stability condition for SFC with multiple spectral crossings remains to be established. Our initial conditions and the ratio of $\bar{\omega}_{\nu_\alpha}/\mu$ are chosen such that the system starts with strong flavor instabilities in the "nonresonant" regime [38]. While a recent work has demonstrated "bipolar-like" evolution of SFC under weak instabilities in the "resonant" regime [39], it is important to conduct self-consistent global QKE simulations including collisions for SFC that take into account the natural appearance of weak instabilities [40–42] as well as their development in a spatially varying matter background [43–45], to fully assess the evolution and impact of SFC in astrophysical environments. With these in mind, our work represents an important step toward the goal of extending the subgrid models developed for FFC [46, 47] to account for the impact of SFC on global neutrino transport.

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Appendix A: Numerical convergence and resolution

We have examined the numerical convergence of our results with different choices of N_z, N_ω, and N_{v_z}, and show in Fig. 9 comparisons of the deviations of the polarization-vector norms from unity in the neutrino and antineutrino sectors (upper panel), defined by

$$\Delta P(t) \equiv \langle |\,|\mathbf{P}(t,z,\omega,v_z)| - 1|\rangle_{z,v_z,\omega>0}\,,$$
$$\Delta \bar{P}(t) \equiv \langle |\,|\mathbf{P}(t,z,\omega,v_z)| - 1|\rangle_{z,v_z,\omega<0}\,. \tag{A1}$$

We also show the evolution of n_{ν_e}/n_ν and $n_{\bar{\nu}_e}/n_{\bar{\nu}}$ (lower panel) for $(N_z, N_{v_z}, N_\omega) = (1000, 2000, 80)$, $(1000, 600, 200)$, and $(2000, 150, 80)$. These comparisons clearly show no visible differences in n_{ν_e}/n_ν and $n_{\bar{\nu}_e}/n_{\bar{\nu}}$, and $\Delta P(t)$ and $\Delta \bar{P}(t)$ remain $\lesssim \mathcal{O}(10^{-2})$ until the end of the simulation.

While the averaged quantities show nearly perfect agreement for all cases, we caution that following the continuous late-time evolution requires a large N_{v_z} after the system seems to have reached a coarse-grained quasistationary state. This is illustrated by Fig. 10, which shows $P_3(v_z, z)$ for $\omega = 0.05$ at different times for the late accretion case in IO using $(N_z, N_{v_z}, N_\omega) = (1000, 600, 200)$. For this case, the ratios n_{ν_e}/n_ν and $n_{\bar{\nu}_e}/n_{\bar{\nu}}$ have already come very close to their quasistationary values at $t \sim 200$ (see Fig. 3), and they continue decreasing slowly before reaching final values at $t \gtrsim 600$. For evolution at late times, Fig. 10 shows that the coherent pattern developed at $t \sim 200$ gets sheared into narrower and narrower filaments, particularly in the v_z direction; fully resolving this cascade therefore requires a large N_{v_z}.

We note that the behavior of the v_z cascade appears to be case dependent. For instance, the cooling cases and the early accretion case in NO retain a more coherent large-scale structure throughout the entire evolution than the late accretion case in IO discussed above. Interestingly, the cases that suffer from severe v_z cascades seem to be the ones whose $\langle P_3(\omega, v_z)\rangle$ lacks correlation with $W(\omega, v_z)$ discussed in Sec. B. Whether these phenomena are linked by an underlying mechanism requires further investigation beyond the scope of this work.

Appendix B: Coarse-grained asymptotic phase-space distributions

We show in Fig. 11 the final coarse-grained function $\langle P_3(\omega, v_z)\rangle_z$ for all six cases (three phases and two mass orderings). These panels show that while the coarse-grained, angle-integrated energy spectra appear to be rather similar in NO and IO (see Figs. 5 and 6), the underlying full phase-space distributions can be very different.

We have also investigated whether the shape of the unstable eigenmodes leaves any imprints on the final distribution of $\langle P_3(\omega, v_z)\rangle_z$. The green-dashed curves in Fig. 11 show the contours of a function $W(\omega, v_z)$, which

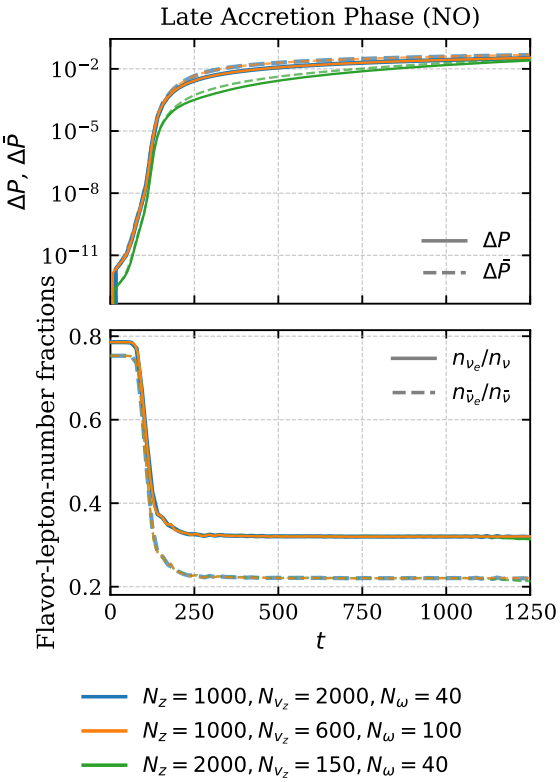


FIG. 9. Evolution of $\Delta P(t)$ and $\Delta \bar{P}(t)$ (upper panel; Eq. (A1)) as well as n_{ν_e}/n_ν and $n_{\bar{\nu}_e}/n_{\bar{\nu}}$ for late accretion in NO using different values of $(N_z, N_{v_z}, N_\omega) = (1000, 2000, 80)$, $(1000, 600, 200)$, and $(2000, 150, 80)$, respectively. t is in units of μ^{-1}.

represents the growth-rate-weighted superposition of the 15 fastest-growing modes among all k obtained in Sec. IV:

$$W(\omega, v_z) \equiv \frac{\sum_{j=1}^{15} e^{\gamma_j t_W}\,|u_j(\omega, v_z)|}{\sum_{j=1}^{15} e^{\gamma_j t_W}}, \tag{B1}$$

where $\gamma_j = \mathrm{Im}\,\Omega_j$ and $u_j(\omega, v_z)$ are, respectively, the growth rate and normalized eigenvector of mode j. For the reference time t_W, we set $t_W = 20\gamma_{\rm max}^{-1}$ using $\gamma_{\rm max}^{-1}$ listed in Table II so that it represents the time when the system has nearly reached the quasistationary state. The numbers attached to the green curves shown in the figure are $\log_{10} W$.

Comparing the shapes of W to $\langle P_3\rangle$ suggests that regions experiencing less flavor conversions (redder) generally have more negative values of $\log_{10} W$. In particular, W qualitatively captures the dominant shape of $\langle P_3\rangle$ for the early accretion case in NO, a part of the phase space in the late accretion case in NO (upper left corner), as well as the two cooling phase cases. However, they do not seem to show any correlations for the two accretion phases in IO. Thus, while W identifies where the instability initially develops, the nonlinear mode–mode coupling

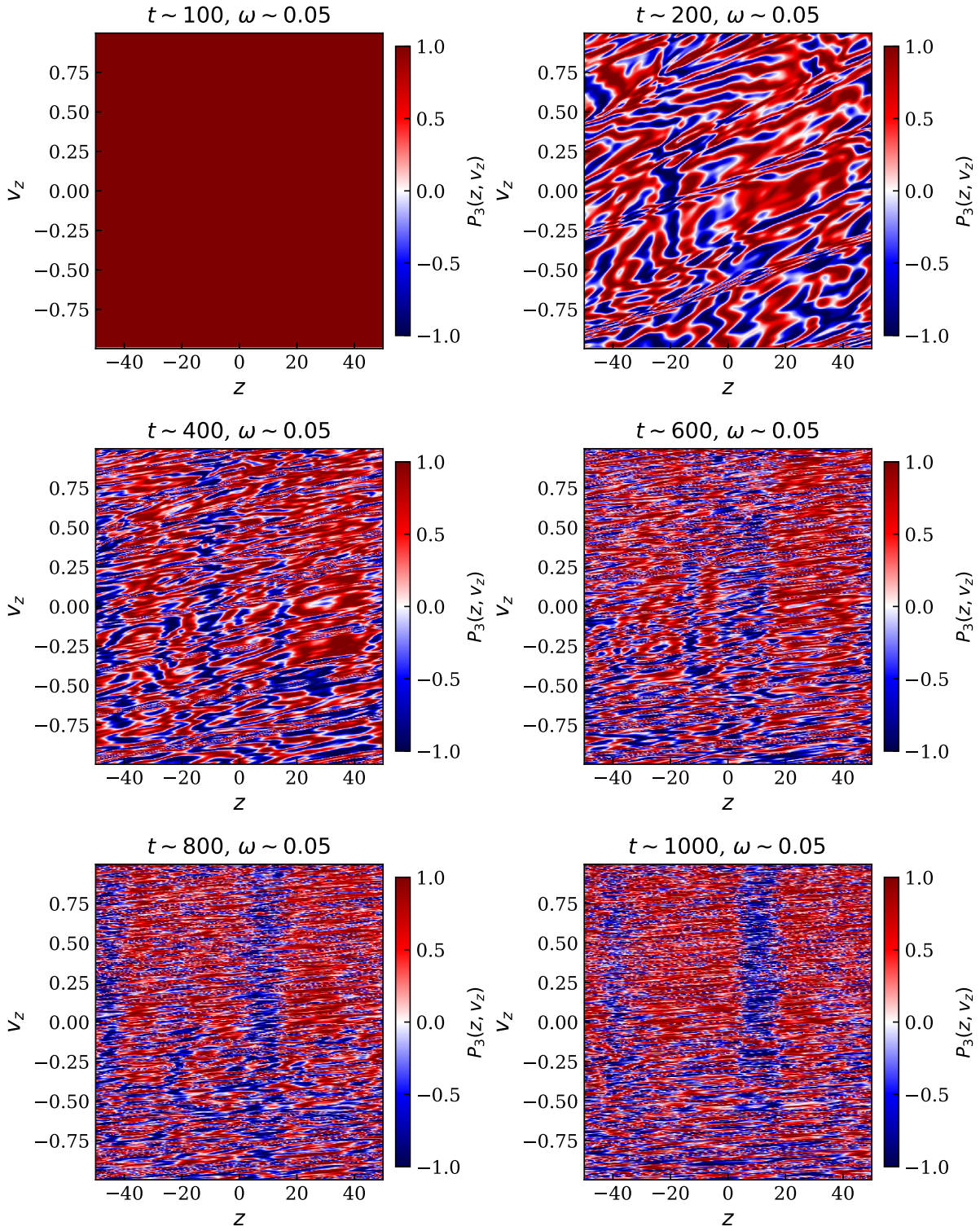


FIG. 10. Snapshots of $P_3(z, v_z)$ for $\omega \simeq 0.05$ for late accretion in IO with $(N_z, N_{v_z}, N_\omega) = (1000, 600, 200)$ at $t \simeq 100, 200, 400, 600, 800$, and 1000. The quantities t and ω are in units of μ^{-1} and μ, respectively.

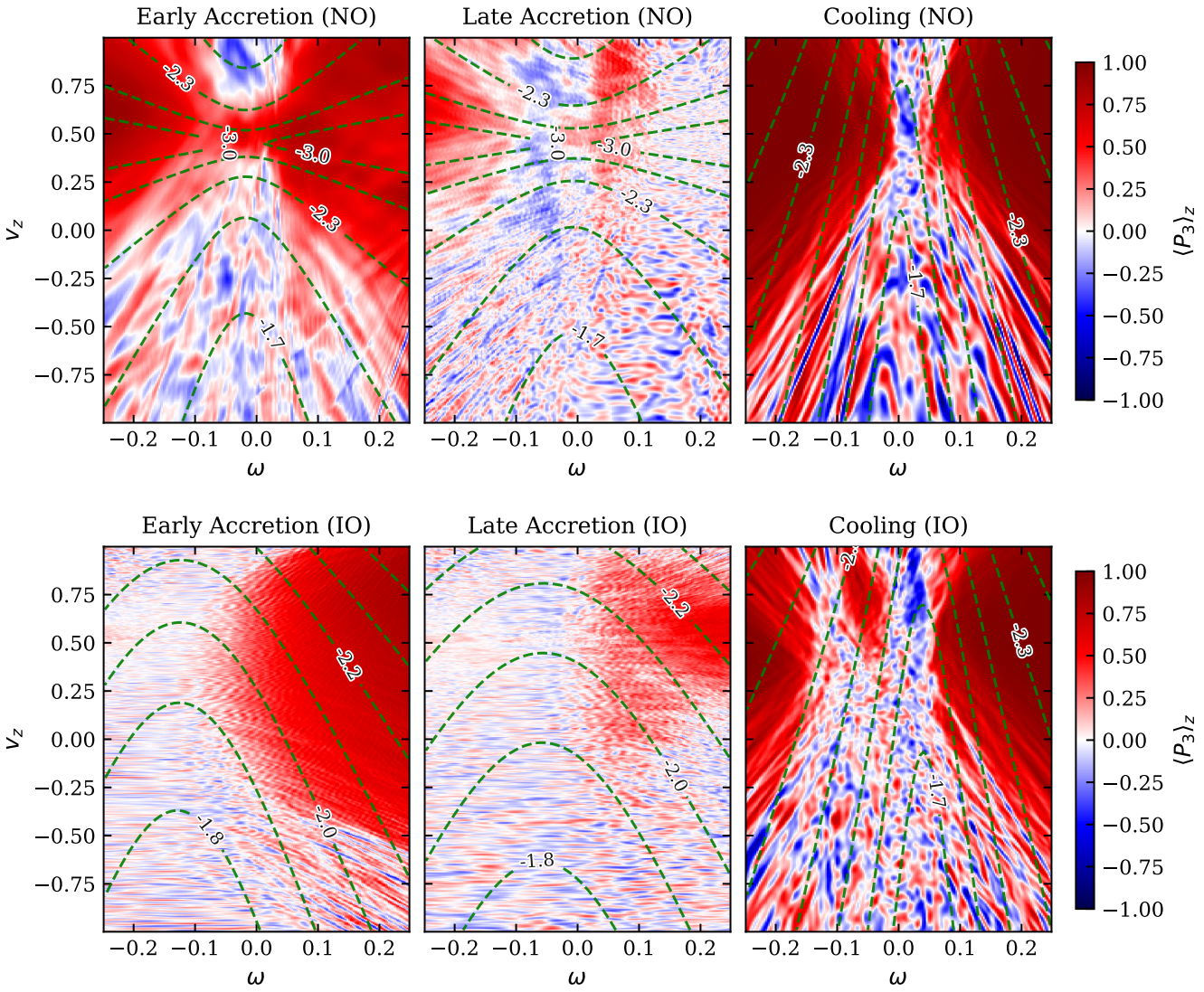


FIG. 11. Spatially averaged polarization vector component $\langle P_3 \rangle_z$ in the (ω, v_z) plane (color) for early accretion (left panels), late accretion (middle panels), and cooling (right panels). The upper and lower rows show NO and IO, respectively, at the final times used in Figs. 7 and 8. Also shown are the contours of $\log_{10} W(\omega, v_z)$ (green dashed) computed using Eq. (B1); see text for details. The quantities t and ω are in units of μ^{-1} and μ, respectively.

may have prevented it from predicting the detailed quasistationary pattern.

Appendix C: Mass ordering dependence

In this appendix, we investigate the insensitivity of the cooling phase flavor evolution to the mass ordering. Figure 12 shows the origin of the weak mass ordering dependence during the cooling phase by resolving the unstable eigenfrequencies into their branches. At the shifted zero mode, $k' = 0$, and writing $v \equiv v_z$ for compactness, the angular kernel is

$$K(v, v') = g_\nu(v')(1 - vv'). \tag{C1}$$

Acting on an arbitrary angular function $u(v)$, this kernel gives

$$(Ku)(v) = \mathcal{A} - v\mathcal{B}, \tag{C2}$$

where

$$\mathcal{A} \equiv \int dv'\, g_\nu(v')u(v'), \qquad \mathcal{B} \equiv \int dv'\, g_\nu(v')v'u(v'), \tag{C3}$$

are simply constants after the integration over v'. Thus, regardless of the original shape of $u(v)$, the result always belongs to the space spanned by 1 and v. Consequently, there can be only two eigenfunctions with nonzero eigenvalues, which we one can write as

$$u_+(v) = a_+ + b_+ v, \qquad u_-(v) = a_- + b_- v. \tag{C4}$$

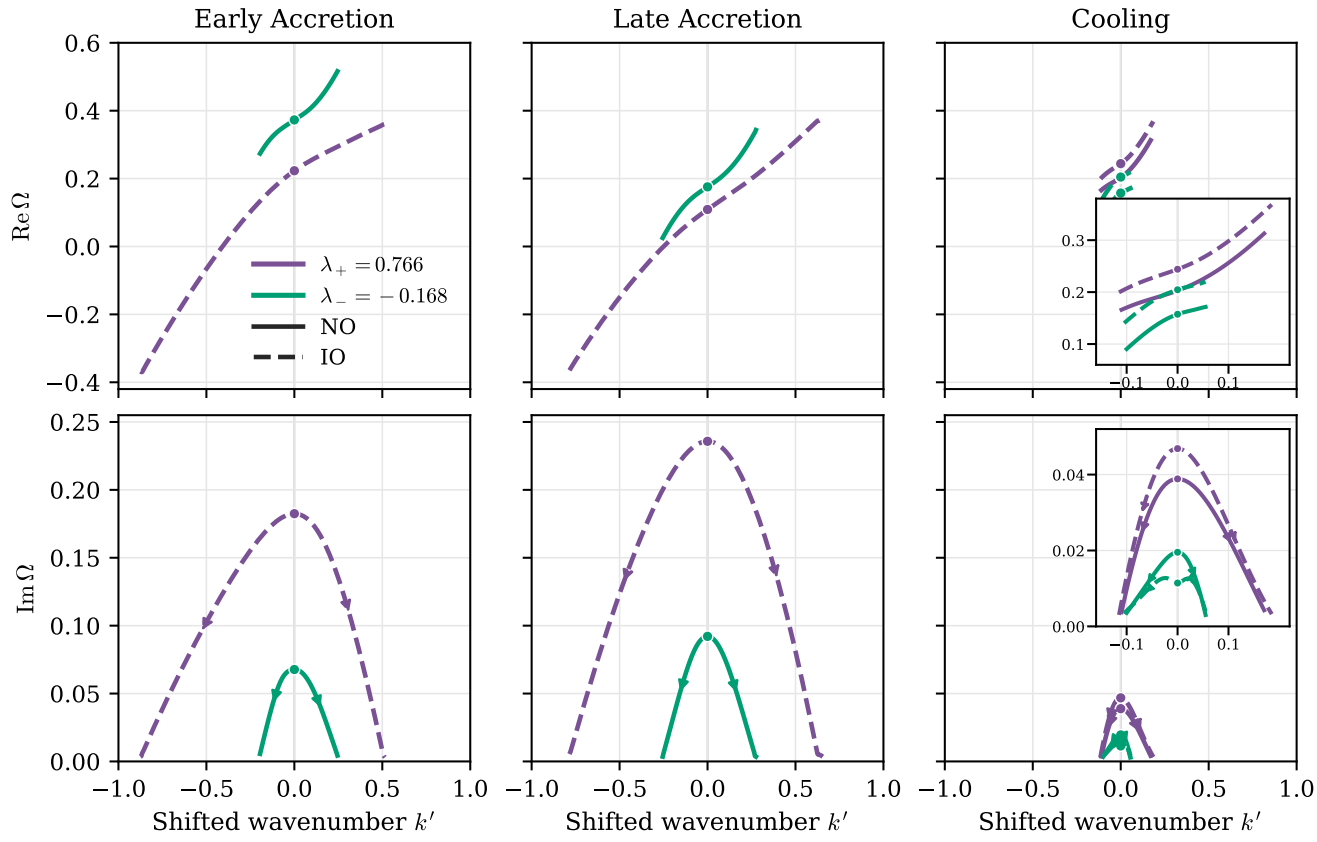


FIG. 12. Real (upper panels) and imaginary (lower panels) parts of the unstable eigenfrequency branches as function of k'. Violet and green label the λ_+ and λ_- eigenvalues (as defined in Eq. C5) that originate at $k' = 0$; solid and dashed curves denote NO and IO mass orderings. Circles mark $k' = 0$, arrows follow the roots toward increasing values of $|k'|$. [**IP:** the main takeway form here is in the ImOmega of the early and late accretion going from NO to IO changes the angualr branch from green to purple. For the cooling phase, going from NO to IO the system maintains the angular branch (purple), see insets.]

Their eigenvalues follow from the moment matrix

$$\begin{pmatrix} 1 & \langle v \rangle \\ -\langle v \rangle & -\langle v^2 \rangle \end{pmatrix} \begin{pmatrix} a_\pm \\ b_\pm \end{pmatrix} = \lambda_\pm \begin{pmatrix} a_\pm \\ b_\pm \end{pmatrix}, \tag{C5}$$

where $\langle v^n \rangle \equiv \int dv\, g_\nu(v) v^n$. For the angular distribution adopted in Eq. (10), this gives $\lambda_+ \simeq 0.766$ and $\lambda_- \simeq -0.168$; the remaining angular functions have zero eigenvalue. For either nonzero mode, the energy-dependent dispersion relation is given by

$$1 + \lambda_\pm \int d\omega\, \frac{G_\omega}{\Omega + \eta_{\rm MO}\omega - \eta} = 0, \qquad \eta = \int d\omega\, G_\omega\,, \tag{C6}$$

where as before, the $\mathrm{Im}\Omega = \gamma$. In the accretion phases, only one angular branch is unstable for each ordering: the instability switches from λ_- in NO to the more strongly coupled λ_+ in IO, explaining the larger IO growth rate. During cooling, both λ_+ and λ_- are unstable, which produces twice as many branches in Fig. 12, while λ_+ remains dominant in both orderings. The mass ordering therefore changes the root along the same dominant branch rather than selecting a different angular mode, so γ changes only from 0.039 to 0.047.

Although not explicitly shown here, we have checked that this cooling phase behavior is robust when varying the width of the angular distribution over $\sigma_\nu = 0.2$, 0.4, and 0.6. We note, however, that this might not be a general feature with arbitrary angular distributions. The dominant mode is λ_+ in both orderings throughout this range in σ_ν, with $\gamma_{\rm NO}(\gamma_{\rm IO}) \simeq 0.026(0.033)$, $0.035(0.043)$, and $0.039(0.047)$, respectively. As σ_ν decreases and the distribution becomes more forward peaked, the weaker λ_- branches progressively disappear: they are unstable in both orderings for $\sigma_\nu = 0.6$, only in NO for $\sigma_\nu = 0.4$, and in neither ordering for $\sigma_\nu = 0.2$. Thus, the exact number of subdominant branches depends on the angular distribution, whereas the weak mass ordering dependence of the dominant cooling instability does not.